

Math 262 – Solutions to Practice Problems
(Every original question and solution; each problem in its own box; condensed to two pages)

Problem 1 — Original Statement and Solution

1. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix}.$$

- (a) Show that T is a linear transformation.
(b) Find the matrix of T .

$$\begin{aligned} a^{(*)} T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) &= T\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}\right) \\ &= \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} - \begin{bmatrix} (x_1 + y_1) + (x_2 + y_2) \\ (x_2 + y_2) + (x_3 + y_3) \end{bmatrix} \\ &= \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} - \begin{bmatrix} x_1 + x_2 + y_1 + y_2 \\ x_2 + x_3 + y_2 + y_3 \end{bmatrix} \\ &= T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) + T\left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\right) \end{aligned}$$

(i) For $r \in \mathbb{R}$

$$T\left(r \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} rx_1 + rx_2 \\ rx_2 + rx_3 \end{bmatrix} = r \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix} = r T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right)$$

So T is linear.

(ii) To find the matrix of T (call it A):

$$\begin{aligned} T(e_1) &= T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ T(e_2) &= T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 + 1 \\ 1 + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ T(e_3) &= T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 + 0 \\ 0 + 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ A &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

2. Given a linear system $A\vec{x} = \vec{b}$, suppose the reduced row echelon form (rref) of the augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Give the general solution to the system.

$$\begin{cases} x_1 + x_4 = 0 \\ x_2 + 2x_4 = 1 \\ x_3 + 3x_4 = 2 \end{cases} \Rightarrow \begin{cases} x_1 = -x_4 \\ x_2 = 2x_4 + 1 \\ x_3 = -3x_4 + 2 \end{cases} \quad \text{letting } x_4 = r$$

$$\text{The general solution: } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -r \\ 2r + 1 \\ -3r + 2 \\ r \end{bmatrix}$$

Problem 3 — Original Statement and Solution

Problem 3 (Original Statement):
Given the linear system

$$\begin{cases} x_1 - x_3 = 1, \\ 2x_1 - 2x_2 = 1, \\ 3x_1 - 4x_2 + x_3 = a, \end{cases}$$

find (if possible) the values of a for which the system has

- (a) no solution,
(b) exactly one solution,
(c) infinitely many solutions.

Also, in the case when there are infinitely many solutions, give the general solution.

Solution:

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ -2 & 2 & 0 & 1 \\ 3 & -4 & 1 & a \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & -2 & 2 & -1 \\ 0 & -4 & 4 & a-3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{2} \\ 0 & 0 & 0 & a-\frac{1}{2} \end{array} \right]$$

(a) No solution if $a \neq \frac{1}{2}$

(b) There is never exactly one solution.

(c) Infinitely many solutions when $a = \frac{1}{2}$

In that case,

$$x_1 = 1 + r, \quad x_2 = \frac{1}{2} + r, \quad x_3 = r \quad (\text{letting } x_3 = r).$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 + r \\ \frac{1}{2} + r \\ r \end{bmatrix}$$

Problem 4 — Original Statement and Solution

In $\mathbb{R}^2$, let

$$\vec{u} = \begin{bmatrix} 1 \\ \frac{1}{\sqrt{10}} \\ 3 \\ \frac{1}{\sqrt{10}} \end{bmatrix}.$$

- (a) Show that $\vec{u}$ is a unit vector.
(b) Let L be the line spanned by $\vec{u}$, and let

$$\vec{x} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

Find the reflection of $\vec{x}$ through L , that is,

$$\text{ref}_L(\vec{x}).$$

Solution:

$$|\vec{u}| = \sqrt{\left(\frac{1}{\sqrt{10}}\right)^2 + \left(\frac{3}{\sqrt{10}}\right)^2} = \sqrt{\frac{1}{10} + \frac{9}{10}} = \sqrt{1} = 1$$

$$\text{proj}_L(\vec{x}) = \frac{\vec{x} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$$

$$\vec{x} \cdot \vec{u} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix} = -\frac{1}{\sqrt{10}} + \frac{12}{\sqrt{10}} = \frac{11}{\sqrt{10}}$$

$$\vec{u} \cdot \vec{u} = 1 \quad (\text{since } \vec{u} \text{ is a unit vector})$$

$$\text{proj}_L(\vec{x}) = \frac{11}{\sqrt{10}} \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} \frac{11}{10} \\ \frac{33}{10} \end{bmatrix}$$

$$\text{ref}_L(\vec{x}) = 2 \text{proj}_L(\vec{x}) - \vec{x}$$

Problem 5 — Original Statement and Solution

Original Problem: In $\mathbb{R}^3$, let L be the line spanned by the vector

$$\vec{w} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Verify that

$$\vec{x} - \text{proj}_L(\vec{x})$$

is orthogonal to $\vec{w}$, that is, their dot product is zero.

Solution:

$$2 \begin{bmatrix} \frac{11}{10} \\ \frac{33}{10} \\ \frac{11}{10} \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{22}{10} \\ \frac{66}{10} \\ \frac{22}{10} \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{22}{10} + 1 \\ \frac{66}{10} - 4 \\ \frac{22}{10} - 4 \end{bmatrix} = \begin{bmatrix} \frac{16}{5} \\ \frac{13}{5} \\ \frac{13}{5} \end{bmatrix}$$

Note: You could also use the matrix formula for this one.

$$\text{proj}_L(\vec{x}) = \frac{\vec{x} \cdot \vec{w}}{\vec{w} \cdot \vec{w}} \vec{w} = \frac{2}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\vec{x} - \text{proj}_L(\vec{x}) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

$$\text{Finally, } \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = 0, \quad \text{so } \vec{x} - \text{proj}_L(\vec{x}) \text{ is orthogonal to } \vec{w} \text{ as claimed.}$$

Problem 6 — Original Statement and Solution

Problem 6 (Original Statement):

Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}.$$

You may assume T is a linear transformation.

- Find A , the matrix representation of T .
- Use A to explain why T is invertible.
- Use A to find the rule for T^{-1} .

Solution:

$$T(e_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

$$\det A = (1)(1) - (-1)(1) = 2 \Rightarrow A \text{ is invertible.}$$

Since $T = T_A$, it follows that T is invertible and

$$T^{-1} = T_{A^{-1}}.$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Thus

$$T^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A^{-1} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(x_1 + x_2) \\ \frac{1}{2}(-x_1 + x_2) \end{bmatrix}.$$

Problem 7 — Original Statement and Solution

Let θ be any angle and let A be the matrix which represents the counter-clockwise rotation through θ . Prove that for any vector

$$\vec{u} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, \quad |A\vec{u}| = |\vec{u}|$$

—that is, rotations preserve length.

Solution:

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad \vec{u} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad A\vec{u} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}.$$

$$|A\vec{u}| = \sqrt{(x \cos \theta - y \sin \theta)^2 + (x \sin \theta + y \cos \theta)^2}$$

Expand:

$$= \sqrt{x^2 \cos^2 \theta - 2xy \cos \theta \sin \theta + y^2 \sin^2 \theta + x^2 \sin^2 \theta + 2xy \sin \theta \cos \theta + y^2 \cos^2 \theta}$$

$$= \sqrt{x^2(\cos^2 \theta + \sin^2 \theta) + y^2(\sin^2 \theta + \cos^2 \theta) + \sqrt{x^2 + y^2} = |\vec{u}|.$$

Hence rotations preserve length.

Problem 8 — Original Statement and Solution

Fill in the blanks for an invertible $n \times n$ matrix A :

- $\text{rank}(A) =$ _____
- The homogeneous system $A\vec{x} = \vec{0}$ has only the _____ solution.
- A is row-reducible to the _____ matrix.

Solution:

$\text{rank}(A) = n$, homogeneous system has only the trivial solution, A is row-reducible to the identity matrix.

Problem 9 — Original Statement and Solution

Let

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 5 \\ 6 & 8 & 4 & 2 & 6 \end{bmatrix}.$$

Give the reduced row echelon form of A .

Solution:

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 2 & 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 & 5 \\ 6 & 8 & 4 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -4 & -1 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 4 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & -4 & -1 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

So the reduced row echelon form of A is

$$\begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Problem 10(a) — Original Statement and Solution

(a) Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Prove that if

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

then

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Solution:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad AB = \begin{bmatrix} a+2c & b+2d \\ 2a+c & 2b+d \end{bmatrix}.$$

If $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, then

$$a + 2c = 0, \quad 2a + c = 0 \Rightarrow a = c = 0,$$

and

$$b + 2d = 0, \quad 2b + d = 0 \Rightarrow b = d = 0.$$

Hence

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Problem 10(b) — Original Statement and Solution

Let

$$C = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

Give the general form of a matrix D such that

$$DC = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Solution:

Let

$$D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then

$$DC = \begin{bmatrix} c & d \\ c & d \end{bmatrix}.$$

For $DC = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, we must have

$$c = 0, \quad d = 0.$$

Hence

$$D = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}.$$

Problem 11 — Original Statement and Solution

In $\mathbb{R}^2$, let

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Show that any vector

$$\vec{w} = \begin{bmatrix} a \\ b \end{bmatrix}$$

can be written as a linear combination

$$\vec{w} = r\vec{u} + s\vec{v}.$$

Solution:

We need to find scalars r and s such that

$$r \begin{bmatrix} 1 \\ 2 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

This gives the system

$$\begin{cases} r + 2s = a, \\ 2r + s = b. \end{cases}$$

Multiply the first equation by 2:

$$2r + 4s = 2a$$

Subtract the second equation:

$$(2r + 4s) - (2r + s) = 2a - b \Rightarrow 3s = 2a - b \Rightarrow s = \frac{2a - b}{3}.$$

Substitute into $r + 2s = a$:

$$r + 2\left(\frac{2a - b}{3}\right) = a \Rightarrow 3r + 4a - 2b = 3a \Rightarrow 3r = 3a - 4a + 2b = -a + 2b$$

Hence

$$\begin{bmatrix} a \\ b \end{bmatrix} = \left(\frac{-a + 2b}{3}\right) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \left(\frac{2a - b}{3}\right) \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Problem 12 — Original Statement and Solution

(a) Is the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

invertible? Why or why not?

(b) The matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

is invertible. Find A^{-1} and use it to solve the equation

$$A\vec{x} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}.$$

Solution:

(a) Augment the matrix with the identity:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & -6 & -12 & -3 & 0 & 1 \end{array} \right].$$

Further row reduction leads to a row of zeros in the left block:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right],$$

so the matrix cannot be reduced to the identity matrix.

Therefore, the matrix is **not invertible**.

(b) Augment A with the identity:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & -2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 1 \\ 0 & 1 & 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 2 & 1 & 1 \end{array} \right]$$

Hence

$$A^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 1 & 2 \\ 2 & -1 & -1 \end{bmatrix}.$$

Solve

$$A\vec{x} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \Rightarrow \vec{x} = A^{-1} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 1 & 2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}.$$

Thus,

$$\vec{x} = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}.$$

Problem 13 — Original Statement and Solution

Problem 13 (Original Statement):

Let

$$A = \begin{bmatrix} 1 & r \\ 2r & 1 \end{bmatrix}.$$

(a) Find $\det A$.

(b) For what value(s) of r is A invertible?

(c) In the cases when A is invertible, give A^{-1} .

Solution:

$$\det A = (1)(1) - (2r)(r) = 1 - 2r^2.$$

$$\det A = 0 \Rightarrow 1 - 2r^2 = 0 \Rightarrow 2r^2 = 1 \Rightarrow r = \pm \frac{1}{\sqrt{2}}.$$

Thus, A is invertible for all $r \neq \pm \frac{1}{\sqrt{2}}$.

$$A^{-1} = \frac{1}{1 - 2r^2} \begin{bmatrix} 1 & -r \\ -2r & 1 \end{bmatrix}.$$